Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

WAVE Thematic Assembly Centre:

WAVE_GLO_PHY_SWH_L4_MY_014_007

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cmems_obs-wave_glo_phy-swh_my_multi-l4-0.5deg_P1D-i

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table. The third column specifies which Sections or Sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	September 2021	All	First version of document	E Charles	E Charles
2.0	June 2024	I.2, I.3, II.4.2, IV, V	Change of the mapping method for VAVH_INST and reprocessing of this parameter	C Béchonnet	P Zunino

LIST OF ACRONYMS

AL	Saral/AltiKa mission
C2	Cryosat-2 mission
CalVal	Calibration/Validation
CF	Climate Forecast (convention for NetCDF)
CFO	CFOSAT mission
CLS	Collecte Localisation Satellite
CMEMS	Copernicus Marine Environment Monitoring Service
DGF	Direct Get File (FTP-like CMEMS service tool to download a NetCDF file)
DT	Delayed Time
DU	Dissemination Unit
EN	Envisat Mission
ERA5	European reanalysis v5
FTP	File Transfer Protocol
GDR	Geophysical Data Record
GOP	Geophysical Ocean Product
HY-2B	Hai Yang-2B mission
IGDR	Intermediate Geophysical Data Record
J1 / J2 / J3	Jason-1 / Jason-2 / Jason-3 missions
L2/L2P/L3/L4	Level 2 / Level 2 Plus / Level 3 / Level 4
LRM	Low Resolution Mode
MIOST	Multiscale Inversion of Ocean Surface Topography
MOTU	Web server allowing the distribution of met/ocean gridded data files through the web (https://github.com/clstoulouse/motu)
NetCDF	Network Common Data Form
NOP	NRT Ocean Product
NRT	Near Real Time
OE	Orbit Error
OGDR	Operational Geophysical data record
PLRM	Pseudo LRM
QUID	QUality Information Document
RMS	Root Mean Square
RPC	Remote Protocol Client
RT	Real Time
S3A	Sentinel-3A mission

S3B	Sentinel-3B mission
SAR	Synthetic Aperture Radar
SARIn	SAR Interferometric
SPC	SPeCtral
Subsetter	CMEMS service tool to download a NetCDF file of a selected geographical box and time range
SWH	Significant Wave Height
TAC	Thematic Assembly Center
TDS	Thredds Data Server
VAVH	Visual AVerage Height
WMS	Web Map Service

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I EXECUTIVE SUMMARY

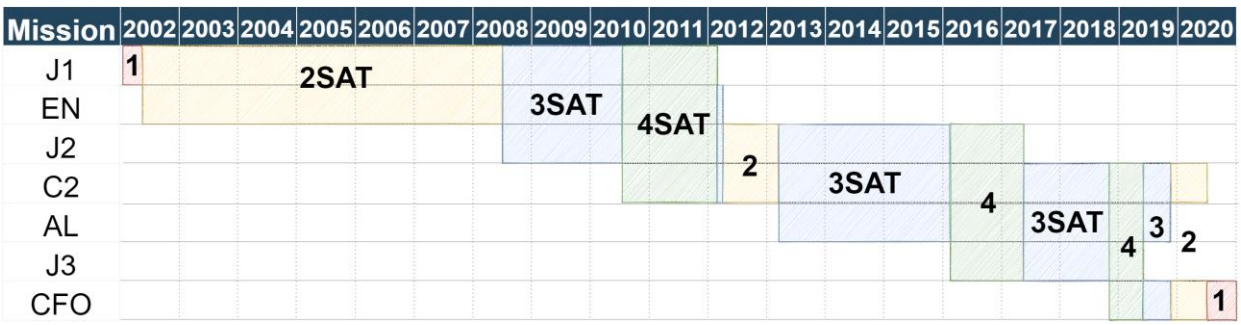
I.1 Products covered by this document

This document describes the quality of the Multi-Year (MY) Level 4 significant wave height (SWH) gridded product listed below:

Product	WAVE_GLO_PHY_SWH_L4_MY_014_007
Area	Global ocean
Missions	Merging of the different altimeter measurements available
Spatial resolution	2°x2° or 0.5°x0.5° Cartesian grid
Temporal resolution	Daily

The number of altimeter data processed by the system varies with time, according to satellites and Multi-Year Level 3 products availability. Table 1 lists the missions used for the mapping of Level 4 SWH product since 2002.

Table 1: Level 3 SWH MY datasets used to compute the Level 4 gridded product. (J1: Jason-1, EN : Envisat, J2 : Jason-2, C2 : Cryosat-2, Al : SARAL/AltiKa, J3 : Jason-3, CFO : CFOSAT) . The numbers correspond to the number of satellites available for processing.



I.2 Summary of the results

The quality of the MY gridded significant wave height (SWH) product was checked upstream by the different providers and was also validated against in-situ measurements.

Comparison with in-situ measurements:

The comparison of the Level 4 gridded product with in-situ data was performed over the 19-year period 2002-2020, with a constellation of one to four satellites (see Table 1).

The Level 4 VAVH_DAILY_MEAN field was compared with daily averaged in-situ data at selected offshore buoys. Computed statistics show a good agreement between daily averaged in-situ data and Level 4 satellite data, with a global root mean square difference (RMSD) of 46 cm, a bias below 1 cm, a scatter index of 15 % and a correlation of 0.93 when considering buoys further than 100 km from the coast.

The Level 4 VAVH_INST field, aiming at reproducing the instantaneous wave field, is compared with in-situ data extracted at 12UTC every day (+/- 3 hours) at selected offshore buoys. Besides the periods with only one satellite, there is good agreement between the buoy data and the VAVH_INST field. In particular, the use of the MIOST mapping method as described in Ubelmann et al. (2021, 2022) as well as the higher resolution, has improved the statistics with a RMSD change from 59 cm to 52 cm, a scatter index change from 23% to 20% (13% improvement) and a correlation increase from 0.9 to 0.92 when considering buoys more than 100 km from the coast. However, the bias has doubled from -3 cm to -6 cm.

For both fields, VAVH_DAILY_MEAN and VAVH_INST, errors are reduced when selecting more offshore buoys.

I.3 Estimated Accuracy Numbers

The EAN are representative of the differences between the Level 4 SWH product and the in-situ measurements. This comparison was performed over the period 2002-2020, with a constellation of one to four satellites (see Table 1). Table 2 summarizes the statistics for the variable VAVH_DAILY_MEAN and the variable VAVH_INST.

Different sets of offshore buoys were selected, depending on their proximity to the coast. Statistics were also computed for several independent buoys, representative of their region.

Table 2: Statistics of comparison of the Level 4 product with in-situ data for selected buoys during the 19-year period 2002 – 2020. The different rows show the results obtained from sets of buoys located 50, 100 and 200 km offshore, and for several individual buoys. Maps were generated from a constellation of one to four satellites. N stands for the number of collocations. RMSD: Root Mean Square Difference; SI: Scatter Index.

Buoys location	VAVH_DAILY_MEAN					VAVH_INST				
	N	Bias (m)	RMSD (m)	SI (%)	R	N	Bias (m)	RMSD (m)	SI (%)	R
Buoys >50 km offshore	336518	0.004	0.47	16	0.93	503119	-0.08	0.54	22	0.92
Buoys >100 km offshore	255913	-0.005	0.46	15	0.93	369641	-0.06	0.52	20	0.92
Buoys >200 km offshore	169215	-0.005	0.44	14	0.94	244484	-0.05	0.50	19	0.93
Buoy 6200001 (Bay of Biscay)	5183	-0.06	0.45	15	0.94	6523	-0.17	0.66	23	0.89
Buoy 41049 (NW Atlantic)	2750	0.03	0.30	12	0.94	4174	-0.04	0.3091	15	0.94
Buoy 6200095 (Western Ireland)	3503	0.13	0.65	15	0.93	4607	0.09	0.73	20	0.91
Buoy 6100002 (NW Med)	403	-0.13	0.58	19	0.83	1092	-0.33	0.85	43	0.70
Buoy 51000 (N Pacific)	2795	0.03	0.36	12	0.89	4025	-0.02	0.42	17	0.86

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production center name

WAVE-CLS-TOULOUSE-FR

II.2 Multi-year system name

The multi-year chain is given a generic name: the L4 alti wave chain.

II.3 Overview of the altimeter-derived SWH measurements

The altimeter sends a spherical radar signal in the direction of the nadir. This signal is reflected by the sea surface and goes back to the satellite. The analysis of the returned signal allows the calculation of the time needed by the signal to go and come back (i.e. the distance between satellite and sea surface). The distribution of the sea state surface elevation impacts the speed at which the return signal is fully returned to the satellite. Hence, the SWH over ocean surfaces is determined from the slope of the front in the radar altimeter wave form. The higher the waves, the more spread is the returned signal in time. Hence, a long delay between the first returns and a full signal return will result in a long shadow in the wave form, which then indicates a high sea state (Figure 1 and 2).

Indeed, when the sea surface is flat, the amplitude of the reflected radar signal increases sharply from the moment the leading edge of the signal hits the surface. However, in a swell or rough sea, the radar signal hits the crest of a wave and then a series of other crests, causing the amplitude of the reflected radar signal to increase more gradually (Figure 1). From the information in this reflected radar signalthe height of the ocean wave can be derived, as the slope of the curve representing its amplitude over time is proportional to the height of the wave (Figure 2).

The term SWH or H_s refers to the mean wave height of the highest third of the waves (also sometimes denoted as $H_{1/3}$).

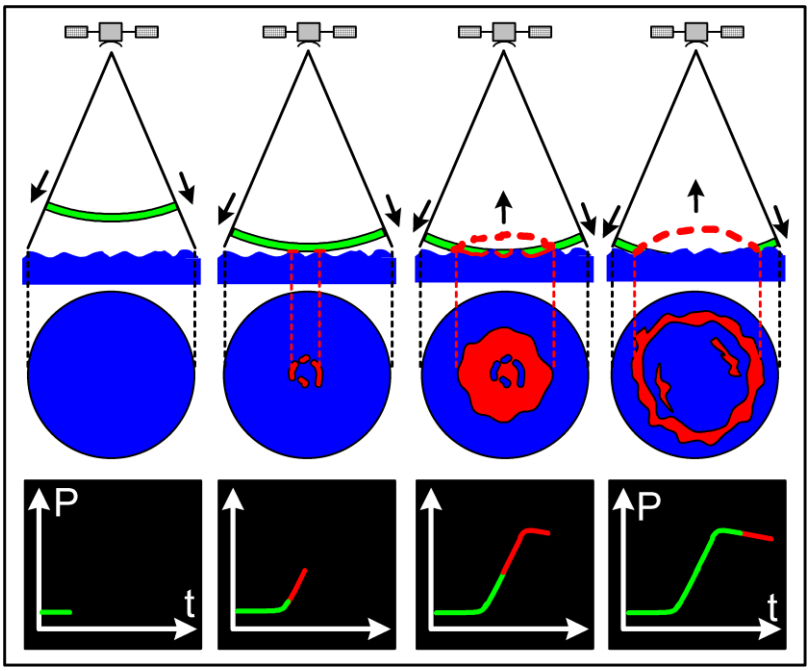


Figure 1: Formation of an echo over a sea surface with waves for conventional altimetry. Top, the satellite, the radar signal emitted by the altimeter (incident: downwards arrows correspond to the green and continuous circle segment, reflected: upwards correspond to the red and dashed circle segment) and the ocean surface seen side-view. Middle, the ocean surface from above, with the radar wave reflection in red. Bottom the radar signal received back by the altimeter (first part of the line, in green: power already received by altimeter, second part of the line, in red: power received at t-time).

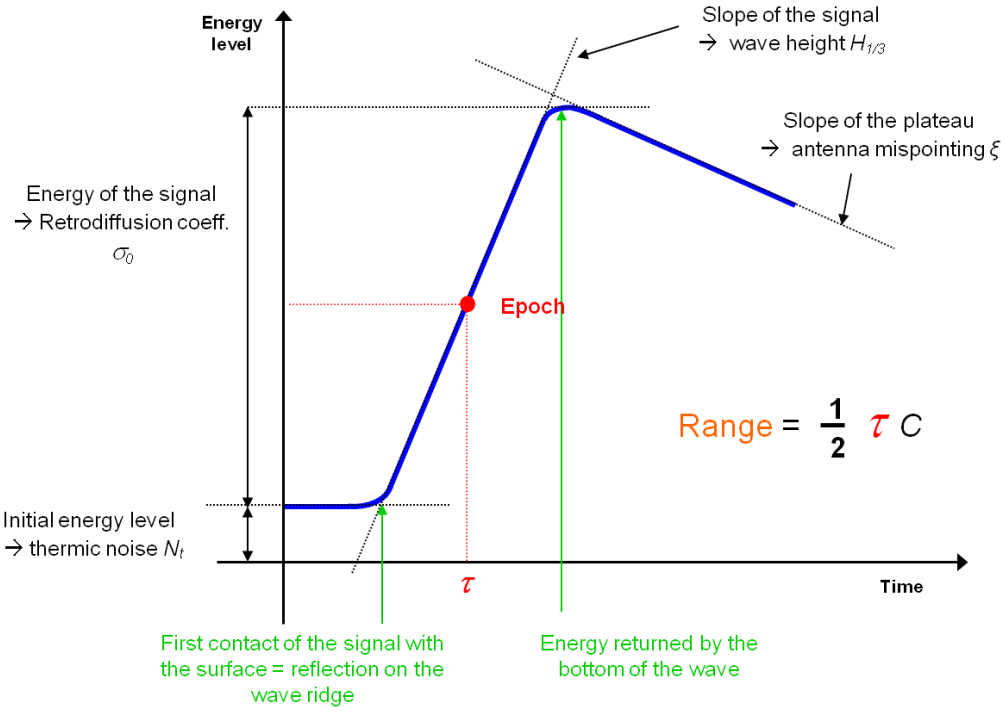


Figure 2: The altimeter waveform.

II.4 Production centre description for the version covered by this document

The system's primary objective is to provide a gridded operational product merging high-quality calibrated significant wave height (SWH) data from all available missions.

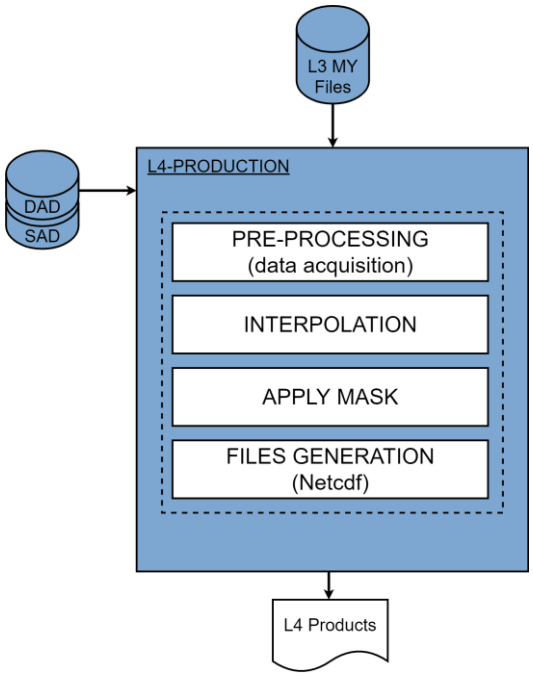


Figure 3: Level 4 Alti wave production component.

The L4 production can be divided into four main steps, illustrated in Figure 3 and described in the next sub-sections:

- Pre-processing: acquisition and selection of upstream data;
- Interpolation: merging and projecting along-track data onto a grid;
- Apply mask: apply a land-sea-ice mask updated daily;
- File generation.

II.4.1 Pre-processing

The altimeter measurements used in the Level 4 system consist of Multi-Year Level 3 products from different missions. These L3 products are in the Copernicus Marine Service catalogue under the name WAVE_GLO_PHY_SWH_L3_MY_014_005. For all missions except CFOSAT, the L3 measurements are derived from ESA Climate Change Initiative (CCI) Sea State project results[1]. CFOSAT Level 3 reprocessing was carried out by the WAVE-TAC from Level-2P reprocessing performed within the CFOSAT project (CNES/CLS). The source and period of availability of L3 products are summarised in Table 3. Mission characteristics are presented in Table 4. Note that for CryoSAT-2, only Low Resolution Mode (LRM)

[1] See CMEMS-WAV-QUID-014-005

measurements are processed, inducing a sparse coverage over the Atlantic and other areas sampled by SAR (Synthetic Aperture Radar) and SARIN (SAR interferometric) instruments.

Upstream L3 products are inter-calibrated with regard to a reference mission and/or in-situ measurements and are filtered using a denoising method based on Empirical Mode Decomposition (EMD) that preserves the meso- and sub-meso scale geophysical signal.

Table 3: Source and period of availability for the different L3 altimeter datasets.

Mission	Type of product	Source	Begin date	End date
Jason-1	L3 MY	ESA CCI (Ifremer)	15/01/2002	03/03/2012
Envisat	L3 MY	ESA CCI (Ifremer)	14/05/2002	08/04/2012
Jason-2	L3 MY	ESA CCI (Ifremer)	04/07/2008	17/05/2017
Cryosat-2	L3 MY	ESA CCI (Ifremer)	16/07/2010	08/07/2020
SARAL/AltiKa	L3 MY	ESA CCI (Ifremer)	14/03/2013	11/11/2019
Jason-3	L3 MY	ESA CCI (Ifremer)	17/02/2016	01/06/2019
CFOSAT	L3 MY	CMEMS	01/11/2018	31/12/2020

Table 4: Altimeter mission characteristics

Altimetric mission	Cycle duration (days)	Latitude range (°N)	Number of tracks per cycle	Inter-track distance at equator (km)	Sun-synchronous	Technology
Jason-1	10	±66	254	~315	No	LRM
Jason-1 Interleaved						
Envisat	35	±81.5	1002	~80	Yes	LRM
Envisat-New Orbit	30		862	-		
Jason-2	10	±66	254	~315	No	LRM
Jason-2 Interleaved						
Cryosat-2	29 (sub cycle)	±88	840	~98	No	LRM only (SAR & SARIN are not processed in MY)
Saral/AltiKa	35	±81.5	1002	~80	Yes	LRM
SARAL-DP/AltiKa	35		1002 (per pseudo-cycle)	-		
Jason-3	10	±66	254	~315	No	LRM
CFOSAT	13	±83	394	-	Yes	LRM

II.4.2 Interpolation

Calibrated and filtered along-track measurements are used to compute two types of gridded fields.

- VAVH_DAILY_MEAN:

This field aims at providing daily statistics derived from satellite measurements on a 2° by 2° grid. Within each grid cell, only measurements of the day (from 00UTC to 23:59 UTC) are used. If no measurements are available in a grid cell, the value is set to NaN. Otherwise, the daily average is computed from available L3 measurements (see Figure 4).

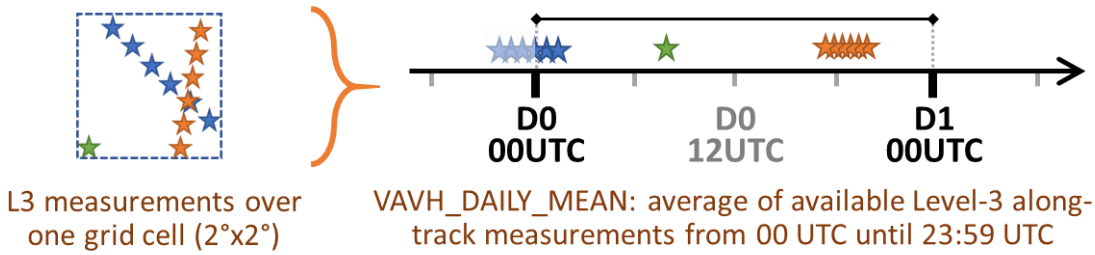


Figure 4: Projection of available measurements on the grid cells and computation of the VAVH_DAILY_MEAN variable.

Additional variables are computed from the available daily L3 measurements: VAVH_DAILY_STD, VAVH_DAILY_MAX and VAVH_DAILY_NBR, respectively the standard deviation, the maximum value, and the number of available measurements.

- VAVH_INST:

This field aims at reproducing the instantaneous wave field at 12UTC with no empty cells. The global SWH maps produced here are based on the Multiscale Inversion of Ocean Surface Topography (MIOST) mapping approach, as described in Ubelmann et al. (2021, 2022). This method is able to account for various modes of variability of the ocean surface topography (e.g., geostrophic, barotrope, equatorial waves dynamic, etc.) by constructing several independent components within an assumed covariance model. The covariance functions in the MIOST system are expressed as wavelet modes, and the inversion is performed in this space using a variational approach. Here, we developed a specific covariance model to represent the evolution of SWH, characterized by spatial scales of 800-1500 km and temporal scales of 1.5 to 5 days. An example of the covariance function for the SWH component is illustrated for a specific location (35°N, 335°E) as a function of space (Figure 5.a) and time (Figure 5.b).

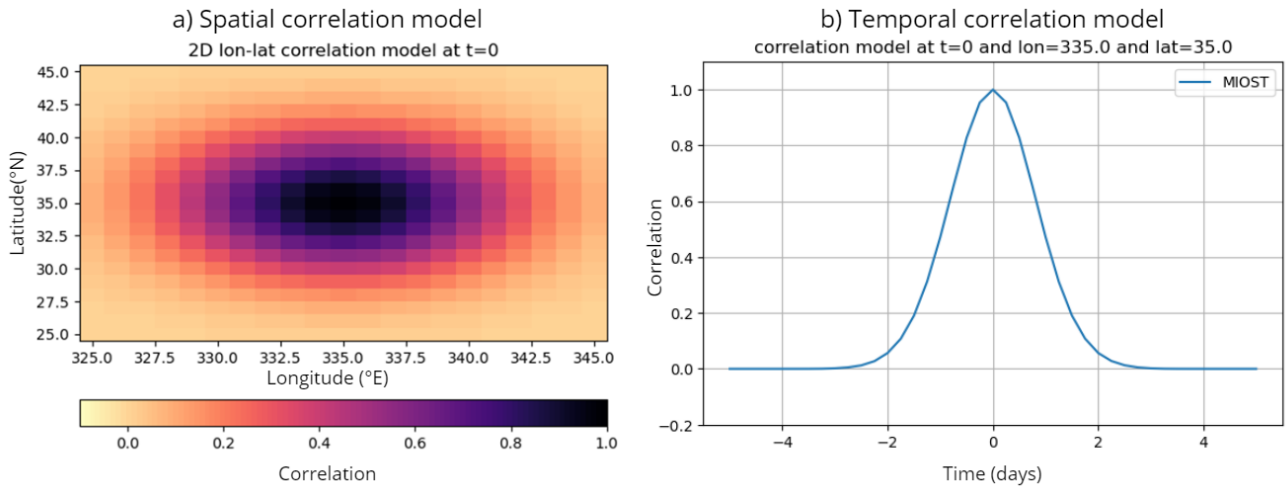


Figure 5: Example of the covariance function for the SWH component used in MIOST at 35°N,335°E, a) as a function of space, b) as a function of time.

An additional variable, VAVH_INST_FLAG is also given to ensure the validity of the data supplied and warn the user about invalid data that has been corrected in post-processing (Table 5). For information, these invalid values correspond to only 0.17% of the entire data set.

II.4.3 Masking

The resulting fields are masked using a static land-sea mask and a dynamic sea-ice mask derived from the CMEMS SEAICE_GLO_SEAICE_L4_REP_OBSERVATIONS_011_009 ice concentration product. Cells are masked when more than a third of the cell is land and when ice concentration is higher than 50%.

II.4.4 File generation

The L4 processing described in previous sections is applied to all available L3 datasets issued from all missions. Two NetCDF files are generated for each day, with time centred on 12UTC: one for the VAVH_INST field and associated variables and the other one for the VAVH_DAILY_MEAN field and associated variables.

The L4 gridded product contains the fields described in Table 5.

Table 5: Gridded significant wave height variables and dimensions included in each L4 NetCDF file.

Name	Standard name *(cell method)*	Long name (units)	Comment
time	time	Time (days since 1950-01-01)	-
latitude	latitude	Latitude (degrees_north)	-
longitude	longitude	Longitude (degrees_east)	-
VAVH_INST	sea_surface_wave_significant_height	Instantaneous significant wave height (m)	The gridded instantaneous significant wave height is estimated from all available altimetry missions Level 3 along-track measurements, accounting for their spatial and temporal proximity.
VAVH_INST_FLAG	sea_surface_wave_significant_height_flag	Flag associated with the modification of the VAVH_INST variable	Flag associated with a post-processing modification of the data (threshold set to 0 m). Where the value is invalid, the flag is set to 1, otherwise to 0
VAVH_DAILY_MEAN	sea_surface_wave_significant_height *(area: mean)*	Daily mean significant wave height (-)	Daily statistics are computed from all available altimetry missions Level 3 along-track measurements from 00 UTC until 23:59 UTC.
VAVH_DAILY_STD	sea_surface_wave_significant_height *(area: standard_deviation)*	Daily standard deviation of significant wave height (m)	
VAVH_DAILY_MAX	sea_surface_wave_significant_height *(area: maximum)*	Daily maximum significant wave height (m)	
VAVH_DAILY_NBR	sea_surface_wave_significant_height number_of_observations	Daily number of significant wave height measurements (-)	

III VALIDATION FRAMEWORK

The validation aims to control the quality of the produced significant wave height and the performance of the key processing steps. Table 6 lists the different metrics that are used to ensure the quality and the integrity of the L4 processing. They cover the following points:

- The number of grid cells defined by the daily maps: Number of grid cells defined by the VAVH_DAILY_MEAN field over the global ocean. This metrics allows to detect potential changes in the input data and therefore in the quality of the mapping.
- The ocean signal temporal consistency: Spatial mean and standard deviation of the L4 VAVH_INST field over the global ocean. This global mean allows to detect potential discrepancies in the mapping method (weighted mean, interpolation, etc.)

The quality control of the upstream along-track data is already performed during the L2P and L3 processing and covers the following points:

- The along-track data availability and spatial/temporal coverage;
- The along-track data editing monitoring;
- The multi-mission cross-calibration monitoring;
- The ocean signal consistency.

Statistics are computed over 30 days.

Table 6: List of the metrics used for WAVE-TAC products operational validation.

Name	Description	Ocean parameter	Supporting reference dataset	Quantity
L4				
SWH_INST_L4-NC-MEAN-GLOBAL	SWH signal monitoring	Significant Wave Height	None	Spatial mean of the VAVH_INST field over the global ocean
SWH_INST_L4-NC-STD-GLOBAL	SWH signal monitoring	Significant Wave Height	None	Spatial standard deviation of the VAVH_INST field over the global ocean
SWH_DAILY_L4-NC-AVAIL-GLOBAL	SWH signal monitoring	Significant Wave Height	None	Number of grid cells defined by the VAVH_DAILY_MEAN field over the global ocean. Statistics are computed over 30 days

IV VALIDATION RESULTS

Validation metrics are used operationally to monitor the quality of the produced significant wave height. These metrics are detailed in section III and are listed in Table 6. In Section IV.1, an example of their application is presented.

A validation study was also performed over the full period, comparing L4 product with in-situ wave data. The results are presented in section IV.2. Also, for the instantaneous variable (VAVH_INST) only, a comparison of the old VAVH_INST and the new VAVH_INST is done by comparison to the reanalyses ERA5 and WAVERYS.

IV.1 Monitoring of the Level 4 significant wave height

Figure 6 and Figure 7 show the validation metrics used to monitor the quality of the produced significant wave height. They are monitored over the full period of the reprocessing to ensure that there are no significant changes in the temporal consistency of the VAVH_INST field and that the spatial coverage of the VAVH_DAILY fields is consistent with the evolution of the constellation. In Figure 6, the decrease in the global mean in 2020 should be related to the quality of the maps produced due to the number of missions available, maximum 2 missions in 2020 (see Table 1); when comparing the product with in-situ data, the differences are larger in 2002 and 2020 (see Table 8) . The variability on the number of cells over time in Figure 7 is also compatible with the number of missions available for the processing (see Table 1).

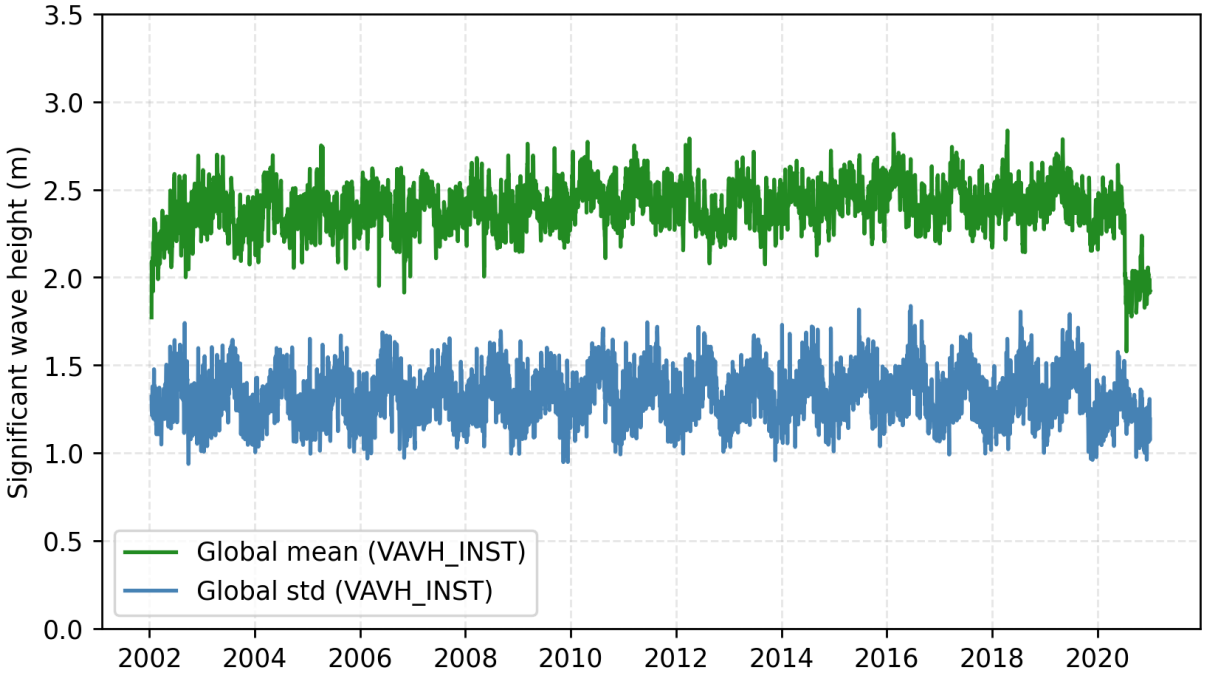


Figure 6: VAVH_INST validation metrics used to monitor the Level 4 reprocessing over the full period (2002-2021).

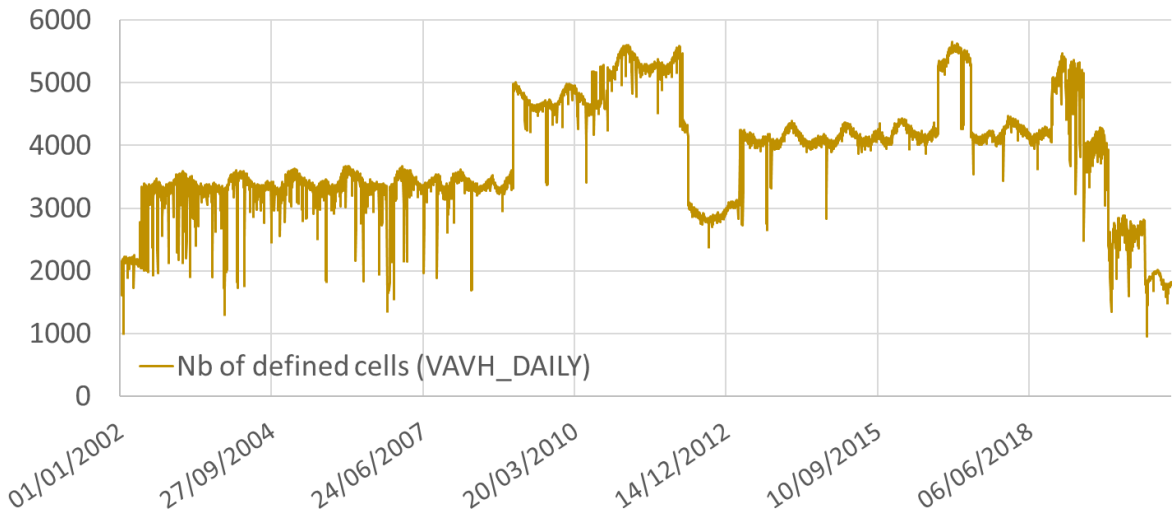


Figure 7: VAVH_DAILY validation metrics used to monitor the Level 4 reprocessing over the full period (2002 – 2020).

IV.2 Validation of L4 SWH product using In-situ data

Different fields are available in the Level 4 gridded products and the validation focussed on the two following fields: the VAVH_INST field and its ability to reproduce instantaneous measurements, and the VAVH_DAILY_MEAN field and its ability to reproduce daily mean values.

This validation is performed over the full period 2002-2020, using Level 4 products merging one to four altimetry missions (Table 1): Jason-1, Envisat, Jason-2, Jason-3, Cryosat-2, SARAL/AltiKa and CFOSAT.

IV.2.1 In-situ data and method

The Level 4 gridded products are compared to available in-situ data issued from the INSITU_GLO_NRT_OBSERVATIONS_013_030 monthly product during the period 2002 – 2020. Only buoys located offshore (at least 30 km away from the coast) and moored in deep water (at least 30 m depth) were selected. The selected buoys are mainly located in the North-East Atlantic Ocean and in the North-West Atlantic and North-East Pacific oceans. They are listed in Appendix A (section 0) and shown in figures 8 to 10.

Valid buoy data were compared with Level 4 data interpolated using inverse distance weighting of the three nearest neighbours. Buoy data were averaged daily to compare with the VAVH_DAILY_MEAN field and extracted at 12UTC every day (+/- 3 hours) to compare with VAVH_INST field.

The following statistics were computed from the daily averaged and the instantaneous timeseries:

- bias
- root mean square difference (RMSD)
- Hannah and Heinold indicator (HH)
- scatter index (SI)
- correlation (R)

The Root Mean Squared Difference (RMSD), defined as:

$$RMSD = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(a_i - b_i)^2}$$

Equation 1

The scatter index (SI), defined as:

$$SI = \sqrt{\frac{\sum_{i=1}^{N}(a_i - \bar{a} - b_i + \bar{b})^2}{\sum_{i=1}^{N} {b_i}^2}}$$

Equation 2

and the Hanna and Heinold indicator (HH, Hanna and Heinold, 1985; see Mentaschi et al, 2013), defined as:

$$HH = \sqrt{\frac{\sum_{i=1}^{N}(a_i - b_i)^2}{\sum_{i=1}^{N} a_i b_i}}$$

Equation 3

where a_i and b_i are the ith colocalised L4 and in-situ observations, respectively, and the overbars represent the mean over all *N* colocalisations. Although the HH indicator is more robust than either the RMSD or the scatter index, the RMSD is easier to interpret physically, and the scatter index is more familiar and more commonly used, and therefore they are both included to aid in comparing with other similar studies.

The results are detailed in the following subsections and main statistics are summarized in Table 7.

Table 7:Comparison Statistics of Level 4 product with in-situ data for selected buoys during the 19-year period 2002 – 2020. The different rows show the results obtained from sets of buoys located 50, 100, 200 and 400 km offshore. L4 variable: "Daily_mean" refers to the VAVH_DAILY_MEAN field (daily mean significant wave height), and "inst" to the VAVH_INST field (instantaneous significant wave height; inst OLD or inst NEW refers to the mapping method used to obtain the field -OLD: 2° x 2° cell, NEW: MIOST, 0.5° x 0.5°). Gridded Level 4 products were generated from a constellation of one to four satellites. "No. of buoys" is the number of buoys used for the validation. "No points" stands for the number of collocations. RMSD: Root Mean Square Difference; SI: Scatter Index; HH: Hannah and Heinold indicator.

Buoys location	L4 variables	No. buoys	No. points	Bias (m)	RMSD (m)	SI (%)	HH (%)	R
Dist >50 km	Daily_mean	160	336518	0.004	0.471	16.4	19.2	0.930
	Inst old	160	502544	-0.006	0.604	24.8	25.5	0.888
	Inst new	160	503119	-0.077	0.536	21.8	22.8	0.915
Dist >100 km	Daily_mean	118	255913	-0.005	0.463	15.4	18.1	0.934
	Inst old	118	369225	-0.032	0.586	22.8	23.5	0.900
	Inst new	118	369641	-0.060	0.523	20.2	20.9	0.922
Dist >200 km	Daily_mean	73	169215	-0.005	0.4375	14.2	16.7	0.939
	Inst old	73	244173	-0.041	0.556	21.1	21.7	0.906
	Inst new	73	244484	-0.049	0.502	19	19.5	0.925
Dis t>400 km	Daily_mean	21	50393	-0.003	0.383	12.2	14.5	0.946
	Inst old	21	73688	-0.019	0.490	18.5	18.9	0.915
	Inst new	21	73761	-0.061	0.458	17.1	17.7	0.928

IV.2.2 Validation of the VAVH_DAILY_MEAN field

The Level 4 VAVH_DAILY_MEAN field is compared with daily averaged in-situ data at selected offshore buoys.

Figure 8 and Figure 9 show the results of the comparison respectively in the European and American regions. Statistics are summarized in the "Daily_mean" rows in Table 7. There is a good agreement between daily averaged in-situ data and Level 4 satellite data, with a global RMSD of 0.5 m, a bias below 1 cm, a scatter index of 16%, a HH of 19% and a correlation of 0.93 when selecting all the buoys located at least 50 km offshore. Errors are reduced when selecting buoys that are further offshore.

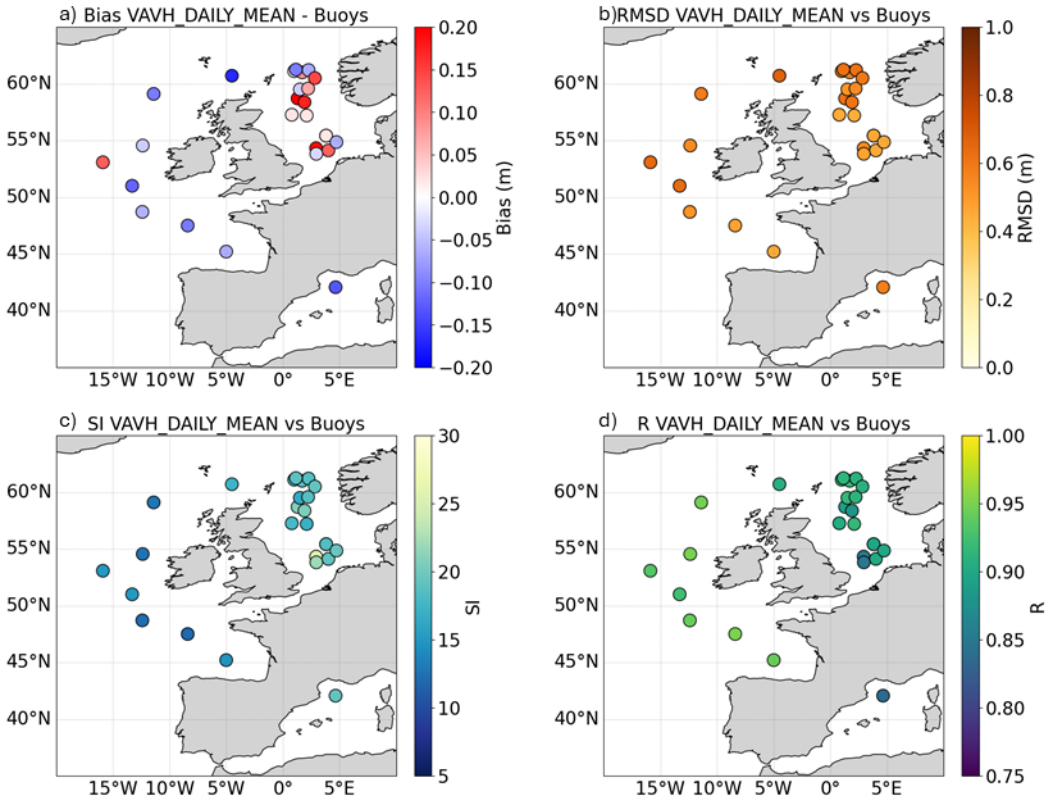


Figure 8: VAVH_DAILY_MEAN field comparison with the network of offshore buoys located in the Western Europe region. a) bias, b) Root Mean Square Difference (RMSD), c) Scatter Index (SI) and d) correlation resulting from the comparison with daily averaged in-situ data during the 19-year period 2002 – 2020.

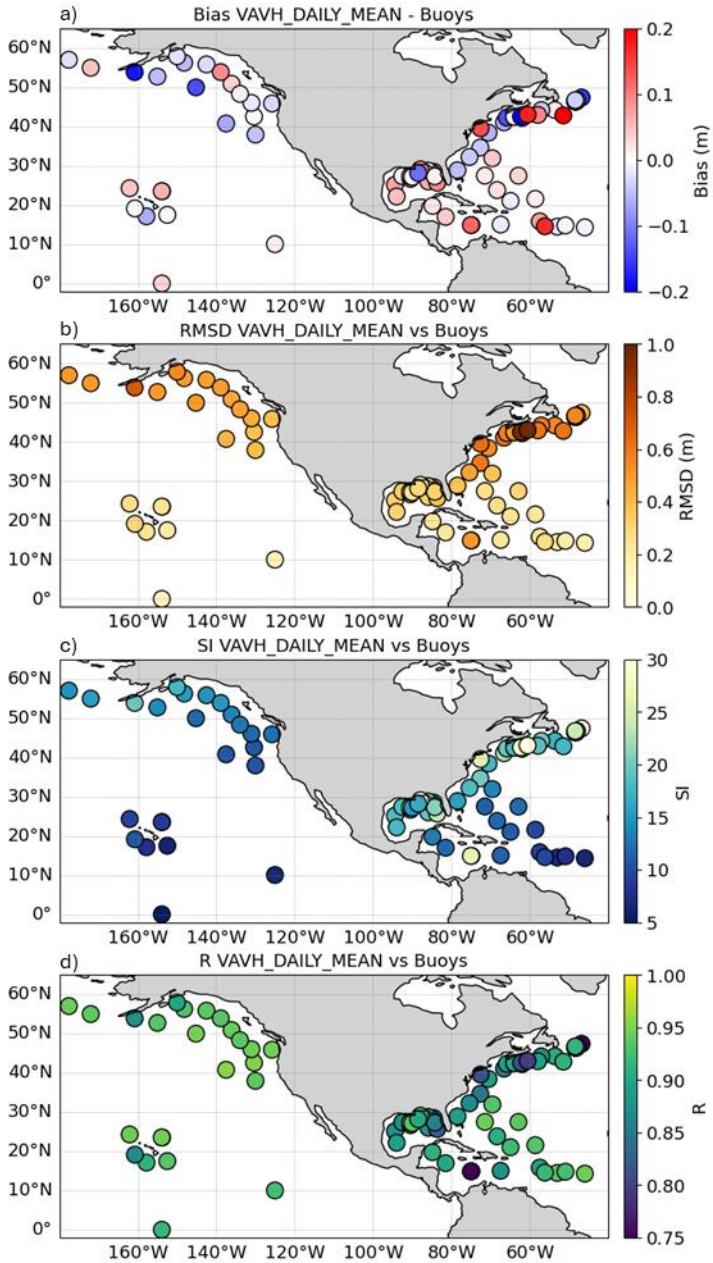


Figure 9: VAVH_DAILY_MEAN field comparison with the network of offshore buoys located in the American region. a) bias, b) Root Mean Square Difference (RMSD), c) Scatter Index (SI) and d) correlation resulting from the comparison with daily averaged in-situ data during the 19-year period 2002 – 2020.

IV.2.3 Validation of the VAVH_INST field

Figure 10 and Figure 11 show the results of the comparison respectively in the European and American regions. The Level 4 VAVH_INST field is compared with instantaneous in-situ data at selected offshore buoys.

Statistics are summarized in the “inst old” and “inst new” rows of Table 7. As the VAVH_INST field attempts to reproduce the instantaneous SWH at 12UTC and is therefore more sensitive to the spatio-temporal coverage of the satellite constellation, the errors are larger than for the VAVH_DAILY_MEAN field. However, there is a still good agreement between in-situ data and Level 4 satellite data and the new mapping method used has improved the RMSD, SI, HH and R, compared with the previous method. Indeed, with a global RMSD of 0.54 m, a scatter index of 22%, a HH of 23% and a correlation of 0.92 when selecting all the buoys located at least 50 km offshore, the performances are better than with the old mapping method. Errors with the new mapping method are further reduced when selecting buoys further offshore, with an overall improvement of about 11% at 100km. Only the bias is degraded by the use of the new mapping method, especially when considering buoys near the coast, but on average, this error remains around -0.06 m.

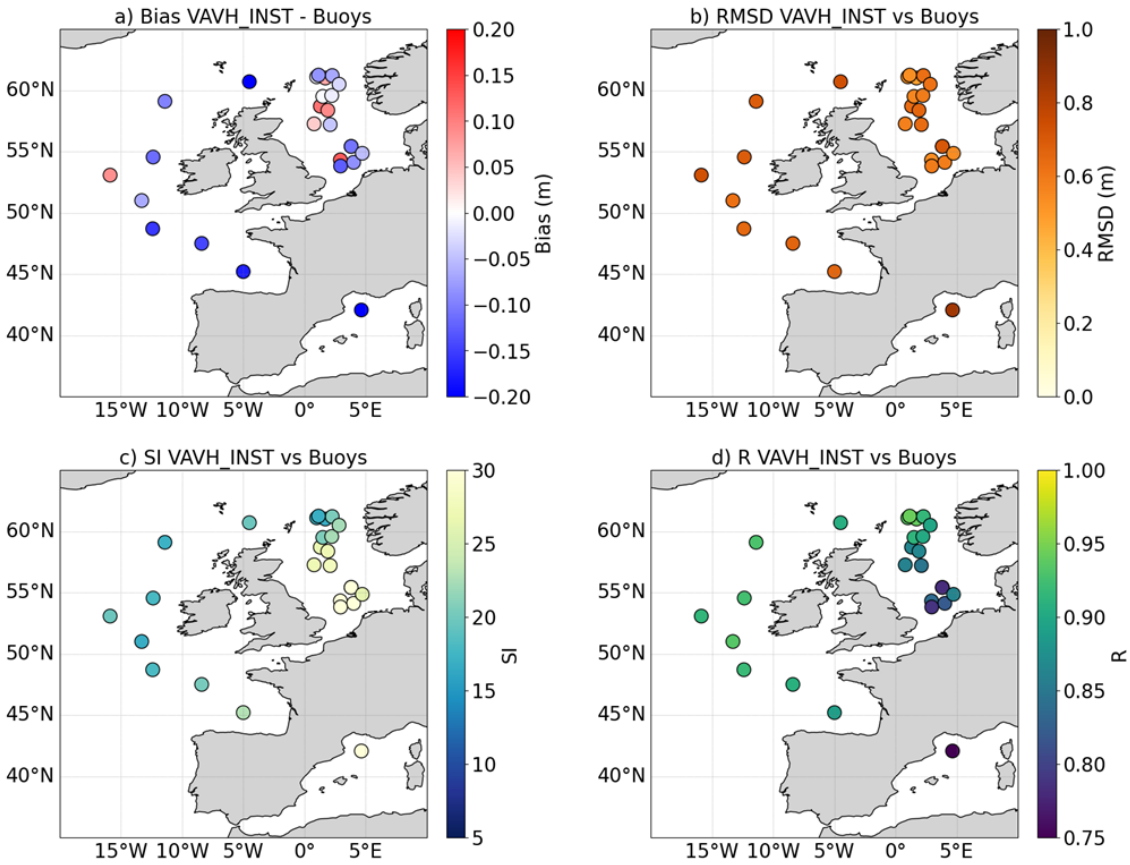


Figure 10: VAVH_INST field comparison with the network of offshore buoys located in the Western Europe region. a) bias, b) Root Mean Square Difference (RMSD), c) Scatter Index (SI) and d) correlation resulting from the comparison with instantaneous in-situ data during the 19-year period 2002 – 2020.

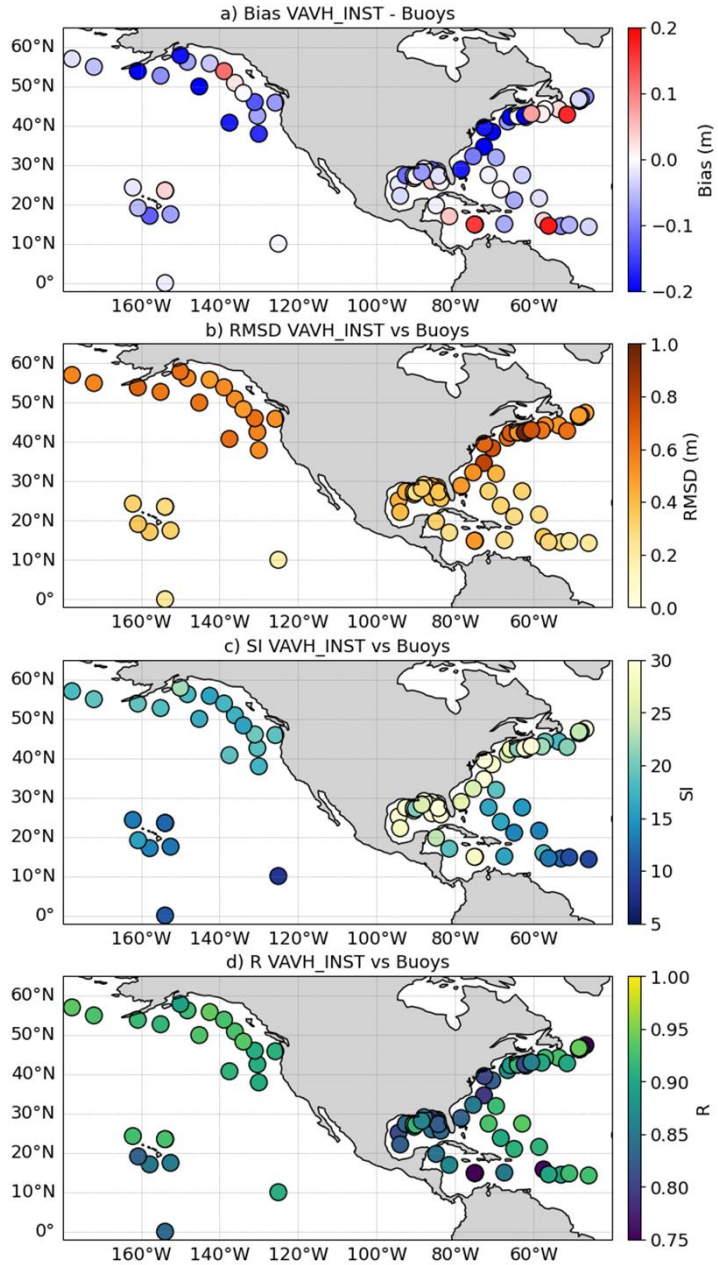


Figure 11: VAVH_INST field comparison with the network of offshore buoys located in the American region. a) bias, b) Root Mean Square Difference (RMSD), c) Scatter Index (SI) and d) correlation resulting from the comparison with instantaneous in-situ data during the 19-year period 2002 – 2020.

IV.2.4 Uncertainty related to the constellation

Over the period 2002 – 2020, numerous constellation events affect the number of available observations to be merged in the maps. Yearly statistics over the whole network of offshore buoys are compared in Table 8 and shown in Figure 12.

As expected, the number of available missions strongly affects the statistics. Statistics are better between 2008 and 2019, when three to four missions were available most of the time. Statistics are lowest in the first and last years of this product especially for the VAVH_INST field, as only one to two missions were available in these years (see Table 1).

Also, the quality and number of in-situ measurements used for this comparison varies over the 2002-2020 period. Therefore, the number of collocated measurements is not the same from one year to another and this also affects the statistics.

Table 8: Statistics of comparison of Level 4 product with in-situ data for selected buoys for each year of the reprocessing. Gridded Level 4 products were generated from a constellation of one to four satellites. N stands for the number of collocations. RMSD: Root Mean Square Difference; SI: Scatter Index, HH: Hannah and Heinold Indicator, R: Pearson Correlation Coefficient.

Year	VAVH_DAILY_MEAN						VAVH_INST					
	N	Bias (m)	RMSD (m)	SI (%)	HH (%)	R	N	Bias (m)	RMSD (m)	SI (%)	HH (%)	R
2002	6484	-0.065	0.508	18.4	18.9	0.933	12135	-0.271	0.912	31.5	35.7	0.811
2003	7541	-0.040	0.512	18.4	18.7	0.931	12033	-0.108	0.570	20.3	21.4	0.919
2004	7985	-0.060	0.499	18.1	18.6	0.933	12537	-0.103	0.537	19.5	20.6	0.926
2005	8758	-0.031	0.499	18.4	18.7	0.930	13831	-0.087	0.542	20.0	20.9	0.921
2006	9334	-0.039	0.503	18.6	18.9	0.933	15236	-0.095	0.541	19.9	20.9	0.926
2007	10205	-0.049	0.505	17.9	18.2	0.937	15824	-0.091	0.534	18.9	19.8	0.933
2008	10599	-0.025	0.484	18.3	18.5	0.932	16642	-0.067	0.520	19.7	20.4	0.925
2009	13918	-0.012	0.469	18.3	18.4	0.935	17826	-0.026	0.449	17.5	17.8	0.942
2010	15013	-0.025	0.440	17.3	17.5	0.936	18362	-0.036	0.421	16.6	17.0	0.945
2011	16975	-0.010	0.429	16.2	16.3	0.945	20025	-0.016	0.427	16.2	16.4	0.949
2012	14369	-0.010	0.454	17.6	17.8	0.933	22942	-0.068	0.480	19.1	19.8	0.927
2013	14610	0.001	0.411	17.0	17.2	0.940	21931	-0.044	0.465	19.1	19.7	0.929
2014	17646	0.014	0.462	18.5	18.6	0.934	24379	-0.016	0.469	18.9	19.3	0.936
2015	19664	0.009	0.453	17.5	17.6	0.938	26883	-0.017	0.465	18.1	18.5	0.938
2016	21121	0.018	0.445	17.3	17.4	0.940	27862	-0.010	0.445	17.5	17.8	0.944
2017	19456	0.030	0.444	17.8	17.9	0.934	24953	0.006	0.422	17.2	17.4	0.942
2018	17082	0.017	0.486	18.9	19.1	0.928	22739	-0.006	0.458	18.1	18.4	0.938
2019	15674	0.022	0.455	18.4	18.5	0.931	22703	-0.045	0.565	22.7	23.5	0.904
2020	9479	0.017	0.449	18.2	18.4	0.932	20798	-0.239	0.767	29.2	33.5	0.840

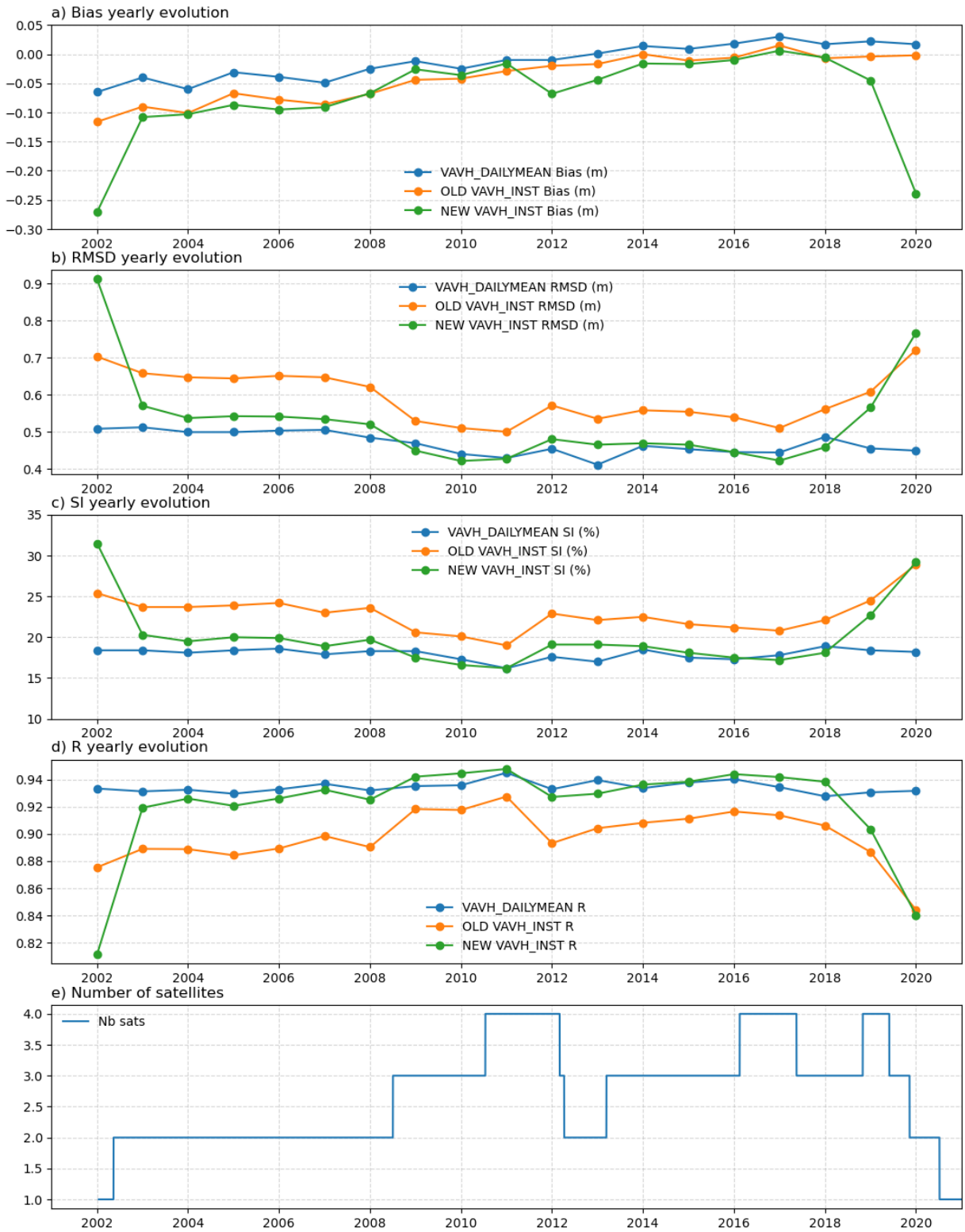


Figure 12: Yearly evolution of the comparison statistics for the Level 4 product with in-situ data in selected buoys. Gridded Level 4 products were generated from a constellation of one to four satellites (panel e).

IV.3 Comparison of new and old L4 VAVH_INST and validation against reanalyses data

The new VAVH_INST results were also visually compared to the old VAVH_INST as well as reanalyses data (ERA5 and the Global Ocean Wave Reanalysis, WAVERYS) to get an overview of the product performance on a global scale. The same statistics as the ones presented in section IV.2.1 were used for a more quantitative comparison of the global VAVH_INST fields.

The WAVERYS SWH and the old VAVH_INST fields are not at the same spatial resolution as the new VAVH_INST and have therefore been interpolated on a 0.5°x0.5° resolution grid. For comparison purposes, all fields were selected at 12:00 UTC.

IV.3.1 Example of old and new VAVH_INST field maps.

By using the new MIOST mapping method, the maps´ resolution has been increased from 2°x2° (left-hand column) to 0.5°x0.5° (right-hand column), giving a smoother, less pixelated effect. However, during periods when only one mission is available (first row), the satellite's path is somewhat more noticeable (Figure 13.b) due to the higher number of interpolations, than with the old version (Figure 13.a). When more missions are available (second row), the maps quality is improved in both cases and especially when using MIOST.

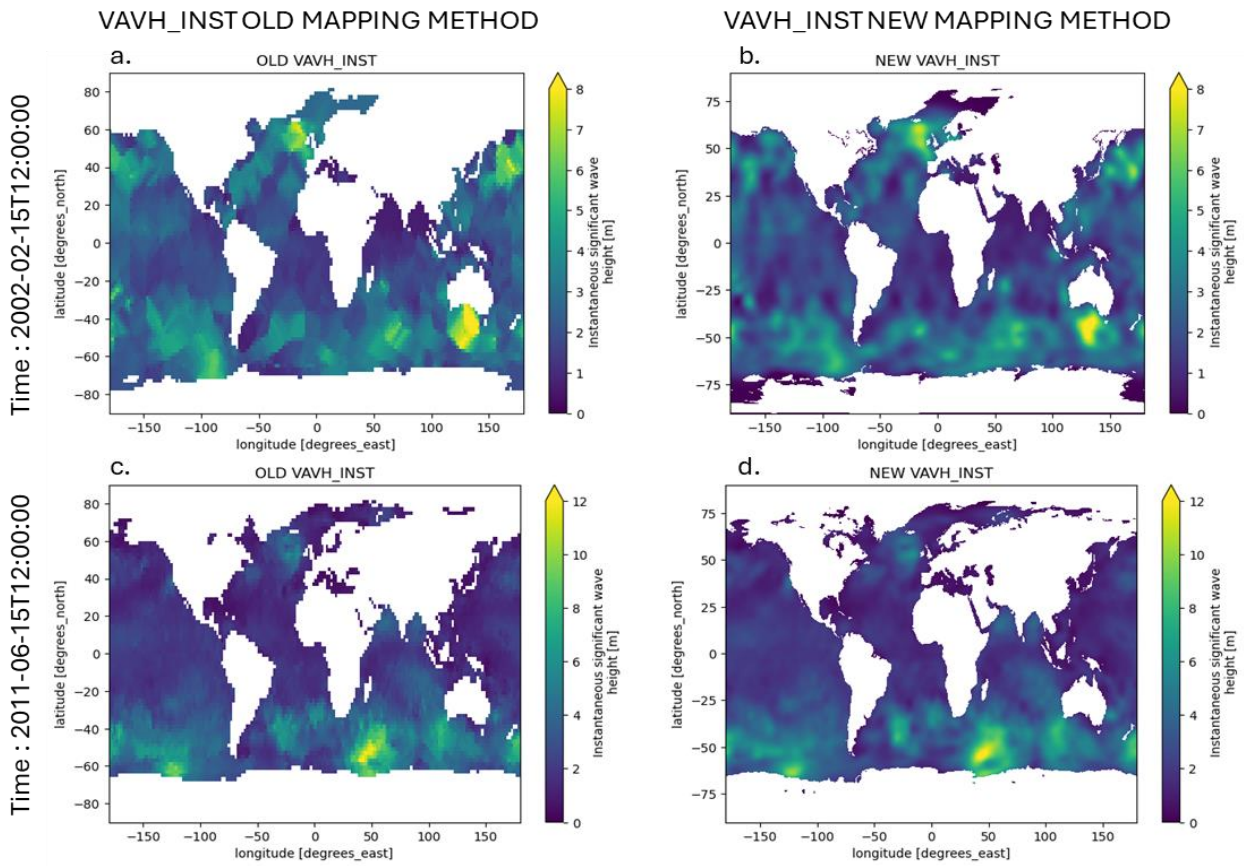


Figure 13: Examples of Level 4 product for two different timestamps (2002-02-15T12h, 2011-06-15T12h) for the old and the new (MIOST) mapping methods. Gridded Level 4 products were generated from a constellation of one (a and b) and four (c and d) satellites.

IV.3.2 Global performance against ERA5 and WAVERYS

Overall, the four maps of SWH averages and variances are fairly similar. They show the same regional patterns in wave heights (Figure 14) and variability (Figure 16) despite the smoothing effect due to interpolation for the old VAVH_INST field.

With the new VAVH_INST field, the SWH average seems closer to the ones of the reanalyses (Figure 14) as it is less smoothed than that of the old VAVH_INST (e.g. in South Atlantic). However, when plotting the relative differences between the VAVH_INST field and the reference (ERA5 or WAVERYS) (Figure 15), between 30°N and 30°S, the mean wave heights of the new field are lower than those of the reanalyses by about 5 to 10%, whereas for the old field the differences vary more between -5 and 5%. Nonetheless, the new VAVH_INST field is closer to the references values in areas where waves are higher (e.g. the South Pacific, Indian Ocean or North Atlantic).

Regarding the variance maps, the one given by the new VAVH_INST field is also closer to those of the reanalyses (Figure 16). Areas of high variability are better represented (e.g. the North Atlantic) and there has been a slight improvement in coastal areas, especially compared to WAVERYS.

Indeed, as shown in Figure 17, the variance of the difference is much lower in areas of high variability (above 40°N and 40°S) when the new MIOST method is used. The new mapping method more closely resembles the behaviour of the reanalyses. However, in areas where there is less variability (such as the equator), there does not seem to be any significant gain from using the new method compared with the previous one.

These results also concur with the correlation maps shown in Figure 18, which reveal a slight increase in correlation in the zones of high variability and minimal gain or even a minor deterioration around the equatorial band.

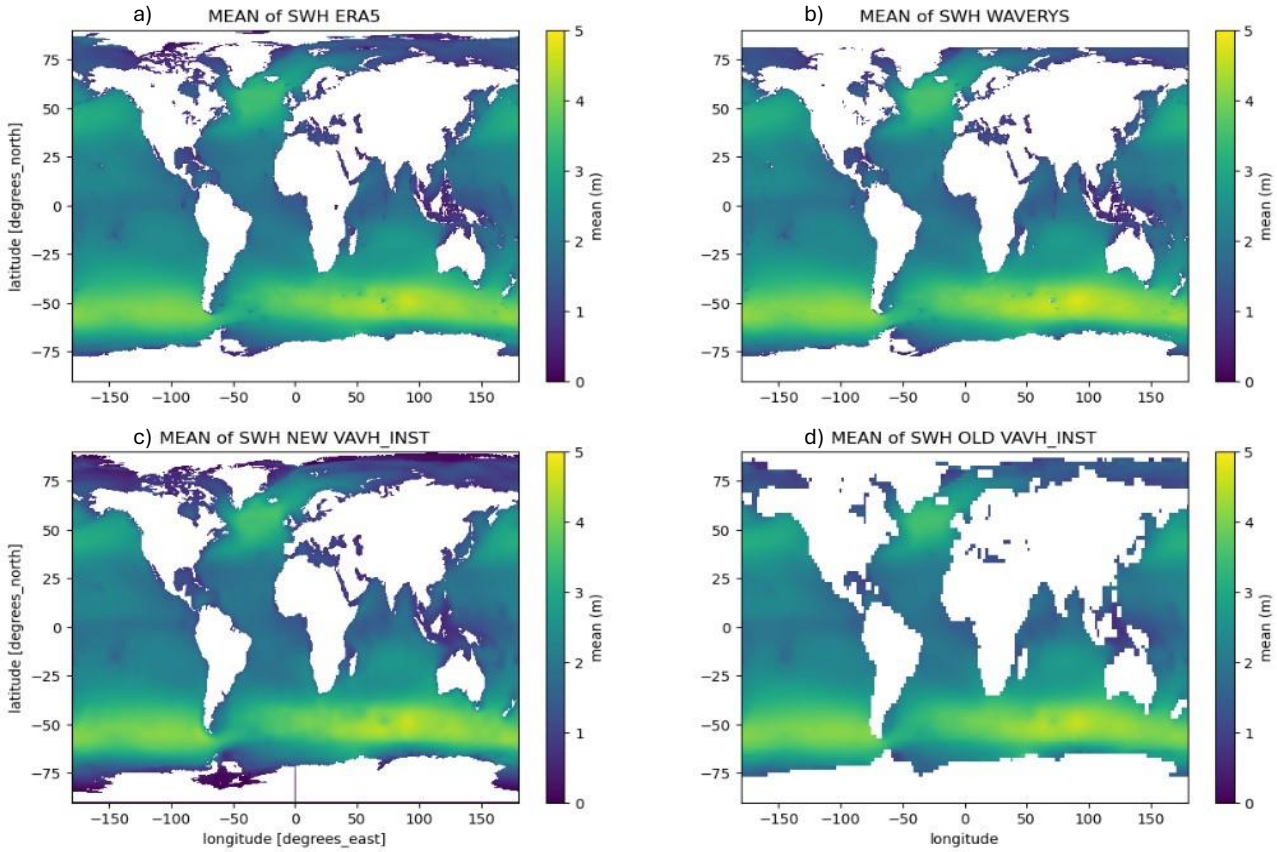


Figure 14: Mean SWH (m) obtained with the four datasets: a) ERA5, b) WAVERYS, c) VAVH_INST created with MIOST (New VAVH_INST) and d) VAVH_INST, created with the old mapping method (Old VAVH_INST).

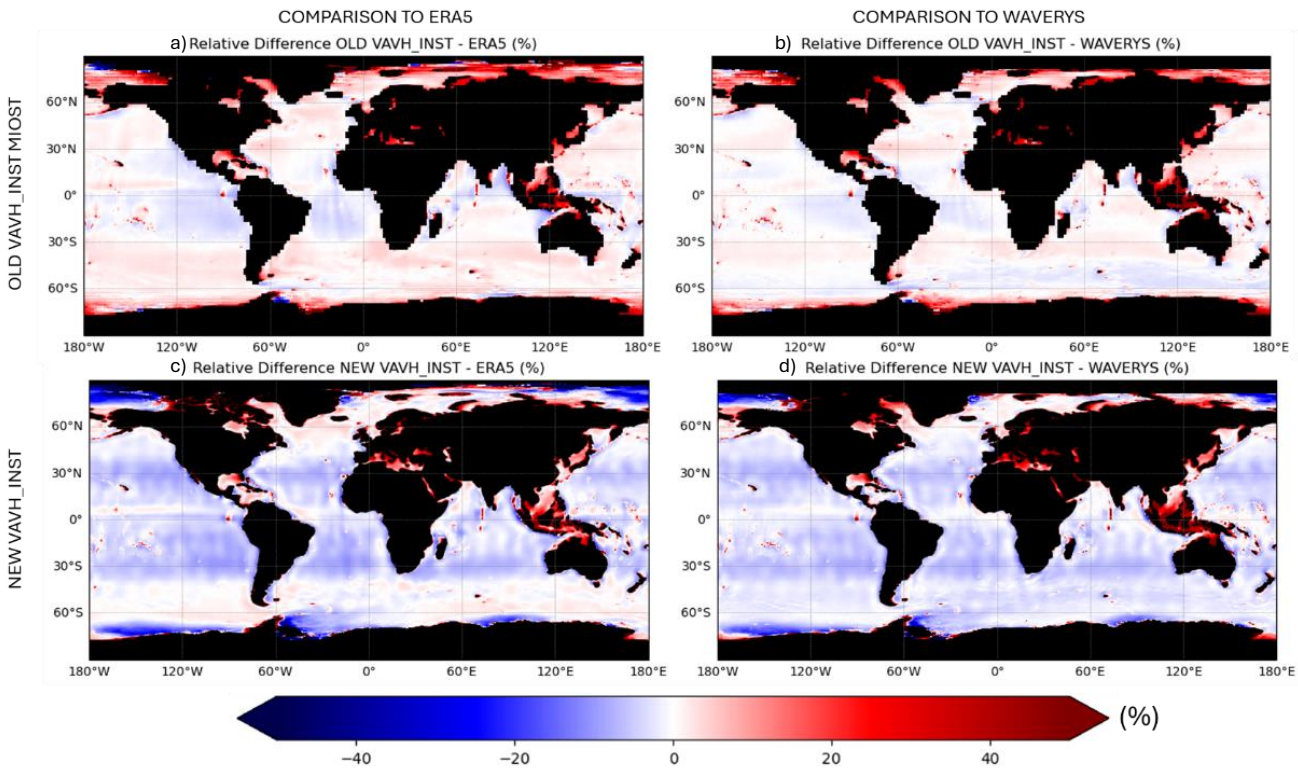


Figure 15: Relative difference in % between VAVH_INST field and the reference (a. OLD VAVH_INST vs ERA5, b. OLD VAVH_INST vs WAVERYS, c. VAVH_INST created with MIOST vs ERA5, d. VAVH_INST created with MIOST vs WAVERYS.

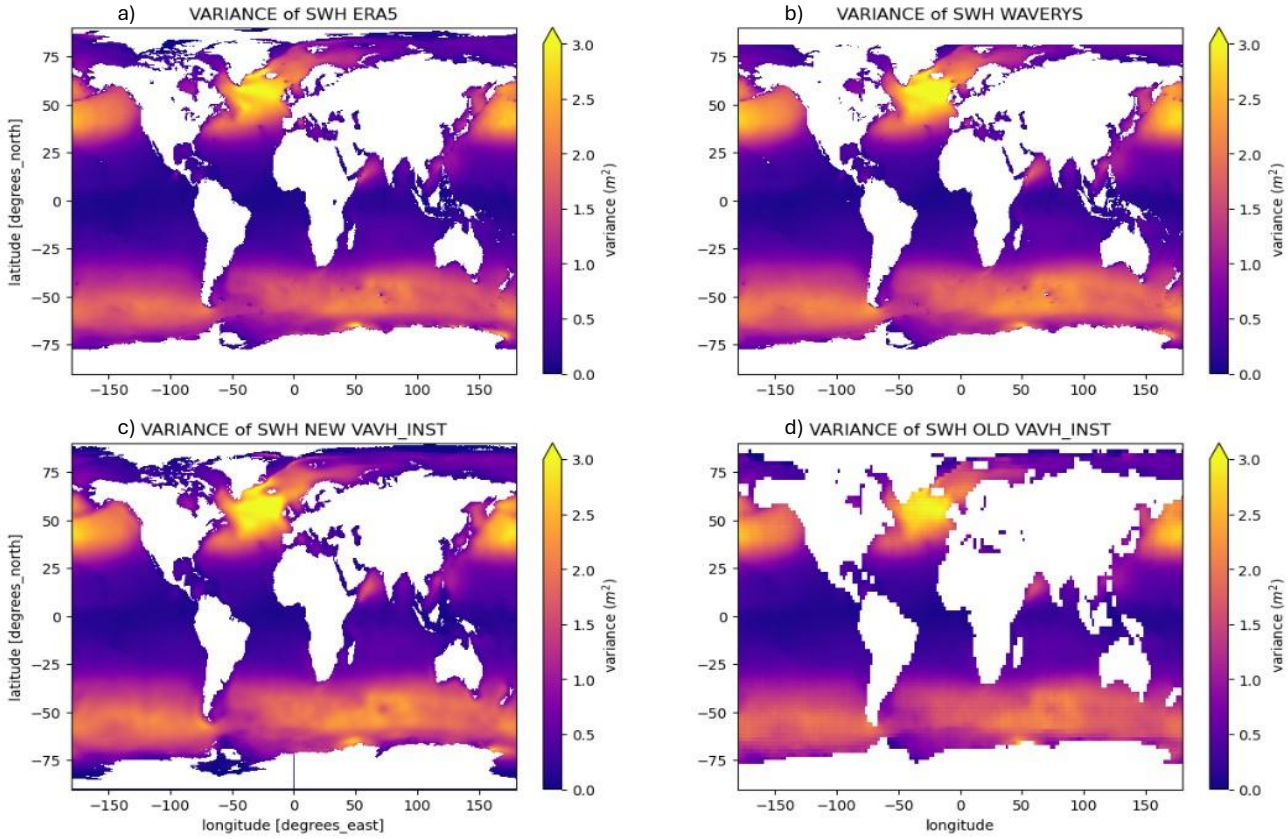


Figure 16: Variance (m) of SWH obtained with the four datasets: a) ERA5, b) WAVERYS, c) VAVH_INST created with MIOST (NEW VAVH_INST) and d) VAVH_INST created with the old mapping method.

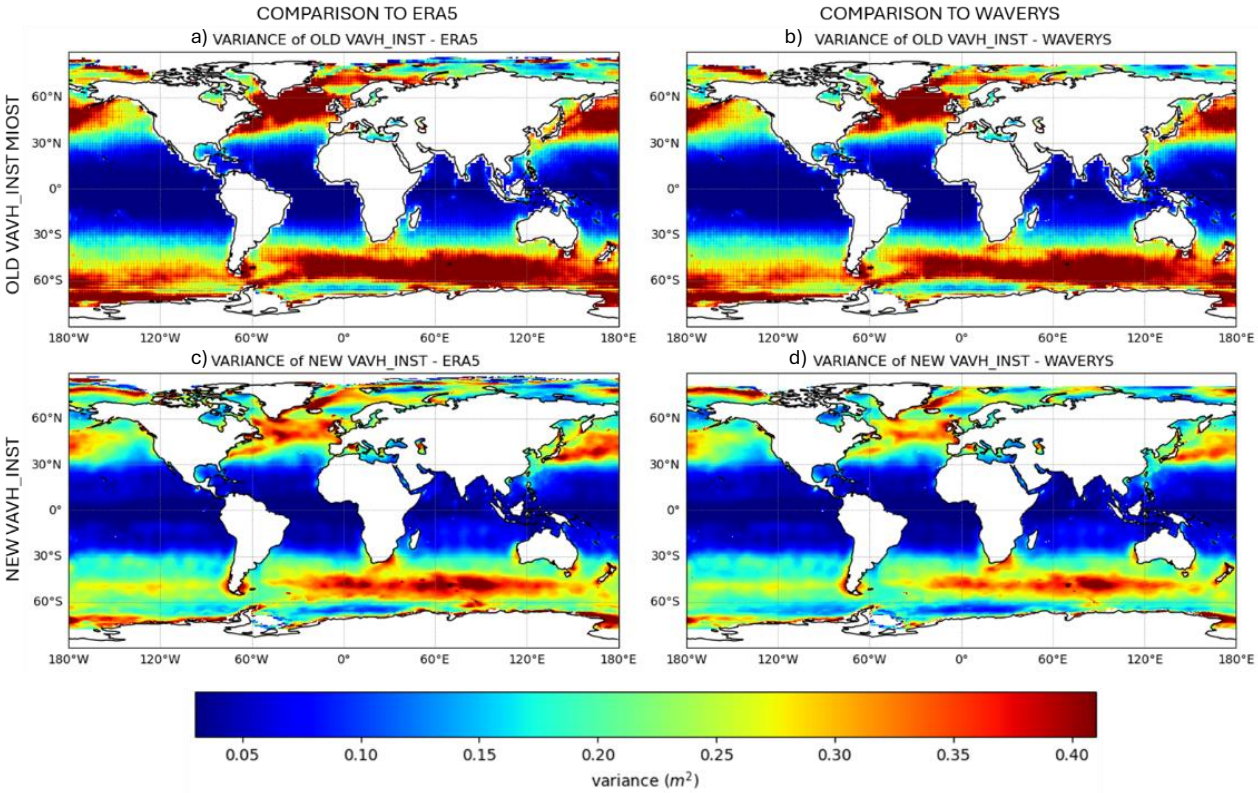


Figure 17: Variance (m2) of the difference between VAVH_INST field and the reference: a) OLD VAVH_INST vs ERA5, b) OLD VAVH_INST vs WAVERYS, c) VAVH_INST created with MIOST vs ERA5, d) VAVH_INST created with MIOST vs WAVERYS.

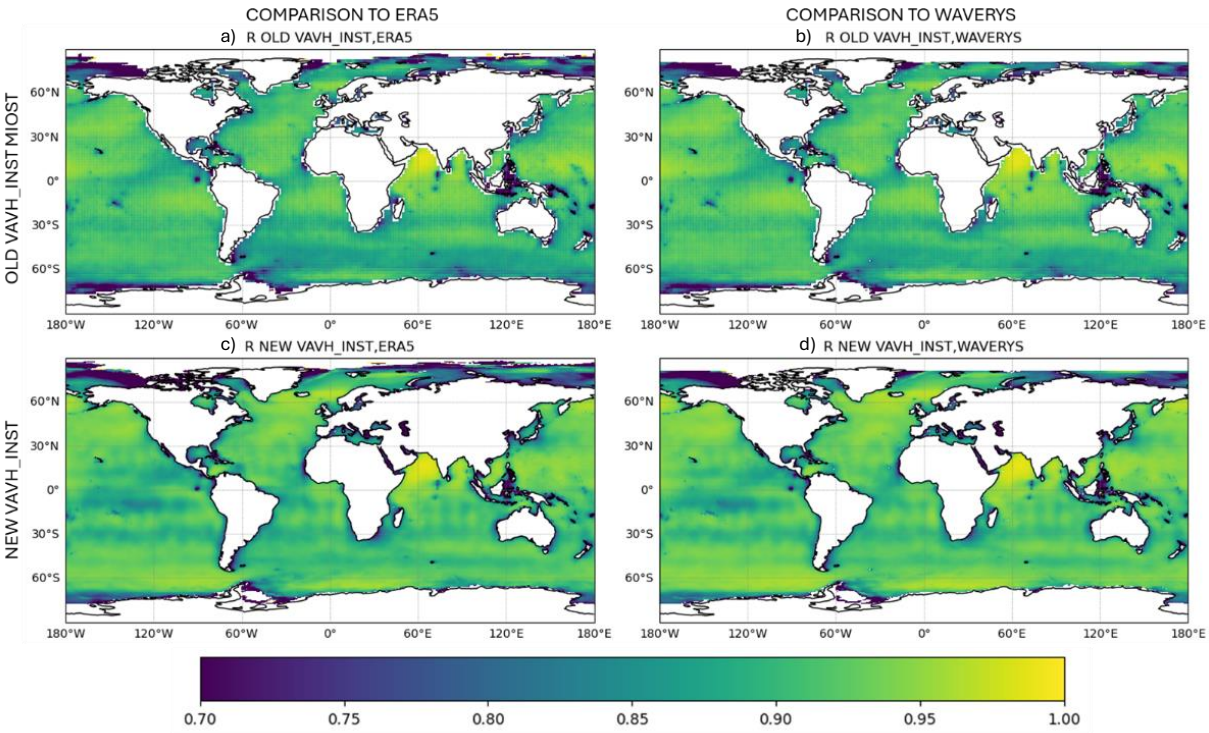


Figure 18: Pearson correlation coefficient (R) between the VAVH_INST field and the reference: a) OLD VAVH_INST vs ERA5, b) OLD VAVH_INST vs WAVERYS, c) VAVH_INST created with MIOST vs ERA5, d) VAVH_INST created with MIOST vs WAVERYS.

IV.3.3 Results Across Oceanic Regime

To better quantify the impact of the new mapping method, this section presents the performance of the VAVH_INST field compared to ERA5 and WAVERYS for different ocean regimes. Only two statistics are presented in the tables 9 and 10: Variance and RMSD of the error between the VAVH_INST field and the reference (ERA5 or WAVERYS).

Overall, the use of the new mapping method has improved performance compared with ERA5 and WAVERYS in offshore regions, particularly in areas of high variability (which is consistent with Figures 16 and 17). For coastal regions, performance compared with ERA5 remains similar regardless of the mapping method, while compared with WAVERYS, there is a tiny improvement in performance with the new mapping method (-4.3% in RMSD and -7.6% in error variance). Finally, around the equator, the error variance and RMSD were already good with the old mapping method. Thus, using the new mapping method only slightly degrades performance relative to the two reanalyses (around 12% in RMSD for ERA5 and 10% for WAVERYS).

Table 9: Variance and Root Mean Square Difference (RMSD) of the mapping error (VAVH_INST maps – ERA5) as a function of oceanic regimes. Coastal: < 200 km; Offshore – High Variability: areas > 200 km + variance > 2 m²; Offshore - Low Variability: areas > 200 km + variance < 2 m²; Equatorial band: between -10°N and 10°N. In orange: the percentage of change relative to OLD VAVH_INST values.

ERA5	VARIANCE (m2)		RMSD (m)	
Oceanic regime	OLD VAVH_INST	NEW VAVH_INST (MIOST)	OLD VAVH_INST	NEW VAVH_INST (MIOST)
Coastal	0.201	0.202 (+0.5%)	0.450	0.450 (=)
Offshore – High variability	0.417	0.306 (-27%)	0.646	0.553 (-14%)
Offshore – Low variability	0.199	0.173 (-13%)	0.446	0.422 (-5%)
Equatorial band	0.029	0.034 (+17%)	0.171	0.191 (+12%)

Table 10: Variance and Root Mean Square Difference (RMSD) of the mapping error (VAVH_INST maps – WAVERYS) as a function of oceanic regimes. Coastal: less than 200 km; Offshore – High Variability: areas > 200 km + variance > 2 m²; Offshore - Low Variability: areas > 200 km + variance < 2 m²; Equatorial band: between -10°N and 10°N. In orange: the percentage of change relative to OLD VAVH_INST values.

WAVERYS	VARIANCE (m²)		RMSD (m)	
Oceanic regime	OLD VAVH_INST	NEW VAVH_INST (MIOST)	OLD VAVH_INST	NEW VAVH_INST (MIOST)
Coastal	0.196	0.181 (-7.6%)	0.444	0.425 (-4.3%)
Offshore – High variability	0.402	0.279 (-31%)	0.639	0.535 (-16%)
Offshore – Low variability	0.195	0.159 (-18%)	0.443	0.411 (-7.2%)
Equatorial band	0.028	0.032 (+14%)	0.166	0.183 (+10%)

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to track and describe changes that may occur in the current multi-year system: due to version or upstream data changes (e.g. addition of a satellite).

V.1 System version changes

The system version changes are summarised in Table 11 below.

Table 11: Multi-Year L4 SWH version changes

System version	Date of Entry in Service of the change	Description of the change	Impact on product quality?
V1.0	DEC-2021	Entry into Service of the Multi-Year L4 SWH product	Yes
V3.0.0	NOV-2024	New mapping method for VAVH_INST multi-Year L4 SWH product and increase of spatial resolution (from 2° to 0.5°)	Yes

VI QUALITY CHANGES SINCE PREVIOUS VERSION

For the VAVH_DAILY_MEAN field, nothing has changed since the previous release.

For the VAVH_INST field, a new mapping method was introduced in this release. This new validation provides a more complete overview of the product's performance (particularly from a global perspective, thanks to the comparison to ERA5 and WAVERYS reanalyses), and shows improvement in the product's quality thanks to the new mapping method: MIOST. Finally, increasing the resolution to 0.5° x 0.5° has particularly enhanced the aesthetics of the maps.

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APPENDIX A: LIST OF IN-SITU BUOYS USED FOR VALIDATION [2002-2020]

In-Situ code	Longitude (°E)	Latitude (°N)	Coastal distance (km)	Number of valid points
2200105	130.000	37.540	67	108698
2300091	89.209	17.810	376	26220
C6S62	124.063	21.677	227	25050
C6V27	118.861	21.021	211	29447
32012	274.922	-19.425	910	83762
41001	287.478	34.502	288	236206
41002	285.165	31.862	340	255258
41004	280.901	32.501	59	218613
41010	281.507	28.872	177	299238
41012	279.466	30.042	70	87112
41036	283.051	34.207	51	78357
41040	306.992	14.477	710	123177
41041	313.907	14.311	1226	113007
41043	295.034	21.061	262	123477
41044	301.386	21.639	546	103483
41046	291.667	23.838	385	114506
41047	288.517	27.517	475	106898
41048	290.503	31.950	436	112916
41049	297.000	27.500	554	105352
41060	308.983	14.824	928	43620
41061	322.000	24.581	1553	9422
41101	303.800	14.600	385	38555
41159	283.052	34.210	51	96509
41300	302.500	15.900	361	19097
42001	270.332	25.897	74	305523
42002	266.334	25.790	323	324975
42003	274.385	25.925	307	309293
42019	264.647	27.913	91	221744
42020	263.306	26.968	67	231465
42036	275.484	28.501	131	204106
42038	267.445	27.421	233	16540
42039	273.992	28.788	113	201452
42040	271.793	29.212	76	187439
42041	269.540	27.500	171	33578
42055	266.000	22.203	220	121658
42056	275.062	19.918	200	111603
42057	278.578	16.908	155	93959
42058	285.082	14.923	293	89805
42059	292.528	15.054	317	96663
42060	296.669	16.433	87	86645

In-Situ code	Longitude (°E)	Latitude (°N)	Coastal distance (km)	Number of valid points
42065	284.954	14.926	289	5032
42097	276.350	25.700	136	82030
42099	275.725	27.345	148	211853
42360	269.541	26.689	122	77250
42369	269.717	27.207	178	23917
42370	269.464	27.321	191	2946
42375	271.711	28.521	95	19210
42376	272.056	29.108	101	3366
42392	269.973	27.196	183	26544
42395	269.208	26.404	94	100973
42887	271.504	28.191	108	23704
44004	289.567	38.484	305	198272
44005	290.872	43.201	69	281126
44008	290.752	40.503	103	281777
44011	293.400	41.105	256	237554
44014	285.160	36.606	94	234694
44024	294.088	42.327	120	82601
44037	292.121	43.491	59	104941
44066	287.382	39.601	123	73053
44088	285.162	36.612	94	42242
44137	298.002	42.262	242	176014
44138	306.367	44.252	262	105661
44139	302.897	44.24	208	159050
44140	308.533	42.868	438	107167
44141	302.042	42.993	182	162714
44142	295.980	42.500	155	81177
44150	295.982	42.505	155	98807
44174	299.143	49.063	58	2220
3100374	314.956	-25.43758	160	4543
3101000	321.569	-3.214	56	8458
4100300	302.510	15.840	355	32183
WEL411	311.567	46.478	337	87420
WEL426	311.563	46.477	337	4991
WEL427	297.711	42.567	230	4245
WEL434	313.163	47.432	437	1724
WEL436	311.566	46.480	337	7984
WEL437	311.567	46.473	337	13914
WEL438	311.566	46.480	337	6047
WEL439	311.563	46.477	337	6765
WEL441	299.818	43.134	89	4097
WEL444	299.396	43.067	106	12838
WEL447	311.810	46.412	357	7834

In-Situ code	Longitude (°E)	Latitude (°N)	Coastal distance (km)	Number of valid points
WEL448	311.570	46.478	337	153996
WEL449	311.989	46.771	360	51129
WEL450	311.547	46.749	328	8882
6100002	4.660	42.060	115	112471
6200001	355	45.200	181	190595
6200029	347.599	48.701	355	189272
6200081	346.699	51	210	183066
6200093	350.001	54.999	71	100334
6200095	344.119	53.075	369	112535
6200105	347.633	54.550	142	174366
6200127	0.700	54	51	41537
6200130	1.300	58.700	192	54855
6200144	1.700	53.400	55	81738
6200145	2.800	53.102	86	83040
6200146	2.100	57.200	236	55396
6200149	1.100	53.700	64	55199
6200163	351.600	47.500	263	205867
6200165	1.100	54	75	43211
6200191	350.42	41.150	70	55835
6200200	351.932	36.398	66	34448
6300110	1.500	59.500	157	80534
6300112	1	61.100	100	82080
6300113	1.708	61	133	64659
6400045	348.599	59.100	214	202896
6400046	355.500	60.701	137	150265
A121	3.817	55.417	243	492371
A122	3.817	55.418	243	241692
Anasuria	0.786	57.260	157	147699
BothnianSea	20.233	61.800	55	107773
D151	2.933	54.317	174	257925
EurogeulDWE	3	51.948	52	419832
F161	4.017	54.117	112	451871
F3platform	4.728	54.854	166	298663
FINO3	7.158	55.195	70	113513
Gullfaks-C	2.269	61.204	122	399212
Heimdal	2.227	59.574	150	360351
J61	2.950	53.817	142	492903
K13a3	3.220	53.218	99	444033
K141	3.633	53.267	76	447955
NorthCormorant	1.149	61.240	111	300967
NsbII	6.333	55	126	62612
Oseberg-A	2.826	60.491	105	385621

In-Situ code	Longitude (°E)	Latitude (°N)	Coastal distance (km)	Number of valid points
Sleipner-A	1.909	58.371	195	368469
SodraOstersjon	18.798	55.920	114	31537
Troll-A	3.719	60.644	52	394655
43010	234.968	10.051	1439	10284
46001	212.080	56.304	289	349212
46002	229.464	42.664	491	286095
46004	223.905	50.93	338	195829
46005	228.999	46.100	492	286956
46006	222.603	40.782	1086	248186
46035	182.250	57.069	443	240696
46036	226.062	48.355	445	229626
46047	240.464	32.403	93	180358
46059	230.049	38.094	536	178039
46066	204.953	52.785	330	134680
46070	175.153	55.082	265	70554
46072	187.912	51.672	70	101353
46073	187.999	55.031	225	77747
46075	199.206	53.969	103	150424
46078	207.360	55.990	96	109955
46080	209.958	57.947	106	98111
46085	217.508	55.868	422	88855
46089	234.181	45.893	140	112680
46098	235.044	44.381	66	36772
46100	235.028	46.851	63	36688
46184	221.150	53.915	348	220260
46205	225.677	54.185	77	210883
46207	230.085	50.875	59	227285
46246	214.800	50.033	921	108162
46259	238.336	34.732	90	93143
48114	190.546	65.012	62	2806
48214	194.752	70.872	131	61907
51000	205.944	23.546	365	97637
51001	198	24.453	152	251191
51002	202.192	17.094	294	258356
51003	199.431	19.289	259	279442
51004	207.618	17.525	338	261599
51028	206.087	0	405	71317
51100	206.100	23.558	376	26739
51101	197.942	24.321	138	101353

APPENDIX B: CHOICE OF THE VAVH_INST VARIABLE TIME WINDOW USED TO SELECT OBSERVATIONS

The gridded VAVH_INST field aims at estimating the instantaneous wave field at 12UTC with no empty cells. Using the MIOST mapping method, all measurements available before and after a given day at 12UTC are considered. These measurements are then projected onto the grid. This temporal window varies spatially to reflect the different temporal decorrelation scales observable for significant wave height and thus, takes into account different wave conditions. The methodology used to obtain the map in Figure 19 is presented hereafter.

Additional metrics were estimated as follow. Several time series of maps with constant decorrelation times were generated: 0.75, 1, 1.5, 3, and 6 days. These time series were then compared with independent along track data (Cryosat-2) to determine the mapping error variance relative to it. Finally, a map of decorrelation time minimising the variance of the mapping error is produced (Figure 19). However, this decorrelation scale map depends on the constellation used.

Values are smaller in areas where wave conditions can change rapidly (e.g. storm affected areas) and larger where wave conditions are steadier (e.g. sheltered areas, equator). The decorrelation time is set at a minimum of 1.5 days and a maximum of 5.7 days.

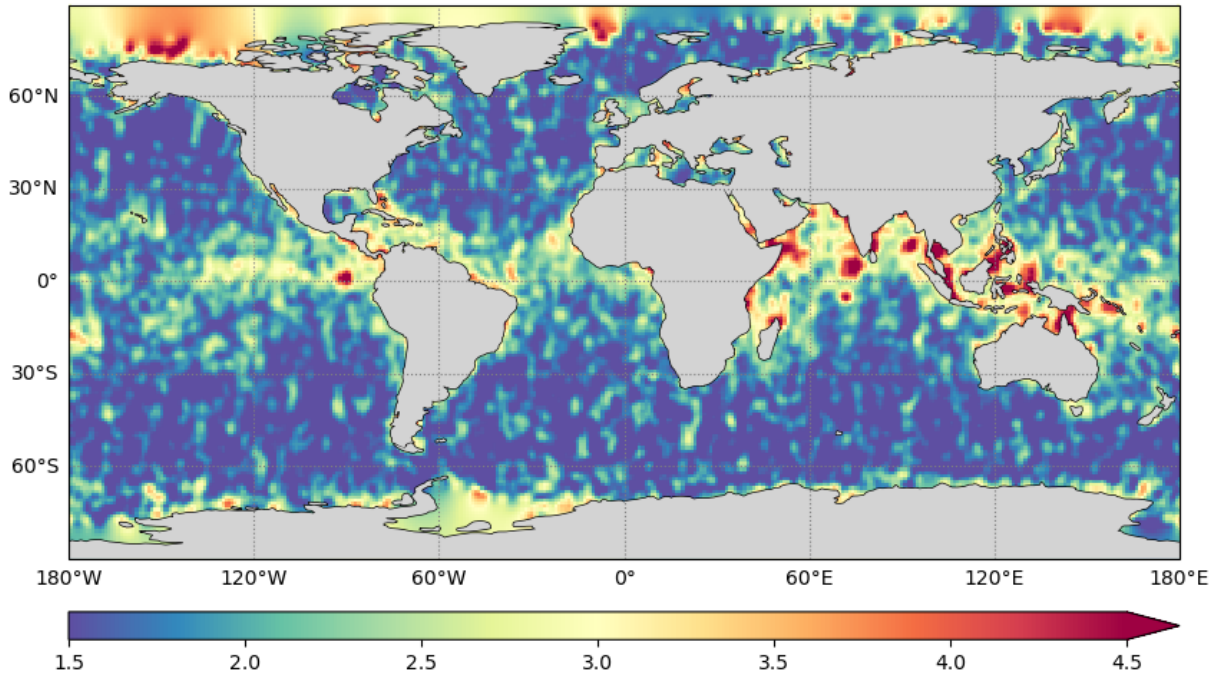

Figure 19: Spatial variability of the decorrelation time (in days) used in MIOST